

Artificial Intelligence (CS311)

Lecture 0: Introduction

Course Information

- Me

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- Main reference:

S. Russel & P. Norvig, Artificial Intelligence – A Modern Approach (3rd Edition)

Please do **not** expect **immediate** response from us, although we will do our best to support your study as much as possible.

Course Requirements - General

- Final Score depends on:
 - final-exam: 40~60 points
 - Homework + attendance: 10~15 points
 - 3 Projects: 30~45 points
- Please **do not negotiate** for more scores (no matter for what reason), unless we make a mistake in calculating your scores.
- Please join the Blackboard site of this course.
- Note: Homework and the 3 projects are **crucial**.
 - e.g., if you never attend and submit only 1 out of 3 projects, you probably will fail.

Course Requirements - Projects

- We can elaborate on the project requirements in lab, but will **not** write a program for you.
- Please finish the 3 projects **independently** (good chance to practice/prepare for your postgraduate study or job).
- Project report/program submitted **after the deadline** will be **marked 0**.
- Discussions and sharing are encouraged, but duplicated submissions, either program or report, will be **marked 0** for all involved submissions.

Outline of this lecture

- What is AI? (state-of-the-art, history, etc...)
- What you can (or cannot) expect to learn from this course?

What is AI? – The Origin

- Most of us believe that human beings are *intelligent*.
- An ambitious question: can the intelligent human beings *build* other entities (agents) that are *at least as intelligent as* we are?
 - What is intelligence and how can we measure it? Sounds like a philosophical question...
 - No unique (neither conclusive) answer.

Everybody could say something about AI

Thinking Humanly “The exciting new effort to make computers think . . . <i>machines with minds</i>, in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)	Thinking Rationally “The study of mental faculties through the use of computational models.” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Acting Humanly “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	Acting Rationally “Computational Intelligence is the study of the design of intelligent agents.” (Poole <i>et al.</i>, 1998) “AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)

Everybody could say something about AI

Thinking Humanly

“The exciting new effort to make computers with minds, in the full sense.” (Haugeland, 1985)

“The study of the [imitation of] activities that we call human thinking, activities such as perception, learning, reasoning, problem solving, learning ...” (Bellman, 1978)



Thinking Rationally

“The study of methods for the use of computational techniques” (Charniak and McMillan, 1985)

“The study of the formal logic of intelligent behavior” (Winston, 1992)



Acting Humanly

“The study of machines that perform tasks that require intelligence as they are done by people.” (Kurzweil, 1990)

“How to make computers do what we call intelligent behavior. At the moment, people are the only machines that do this.” (Ed Knight, 1991)



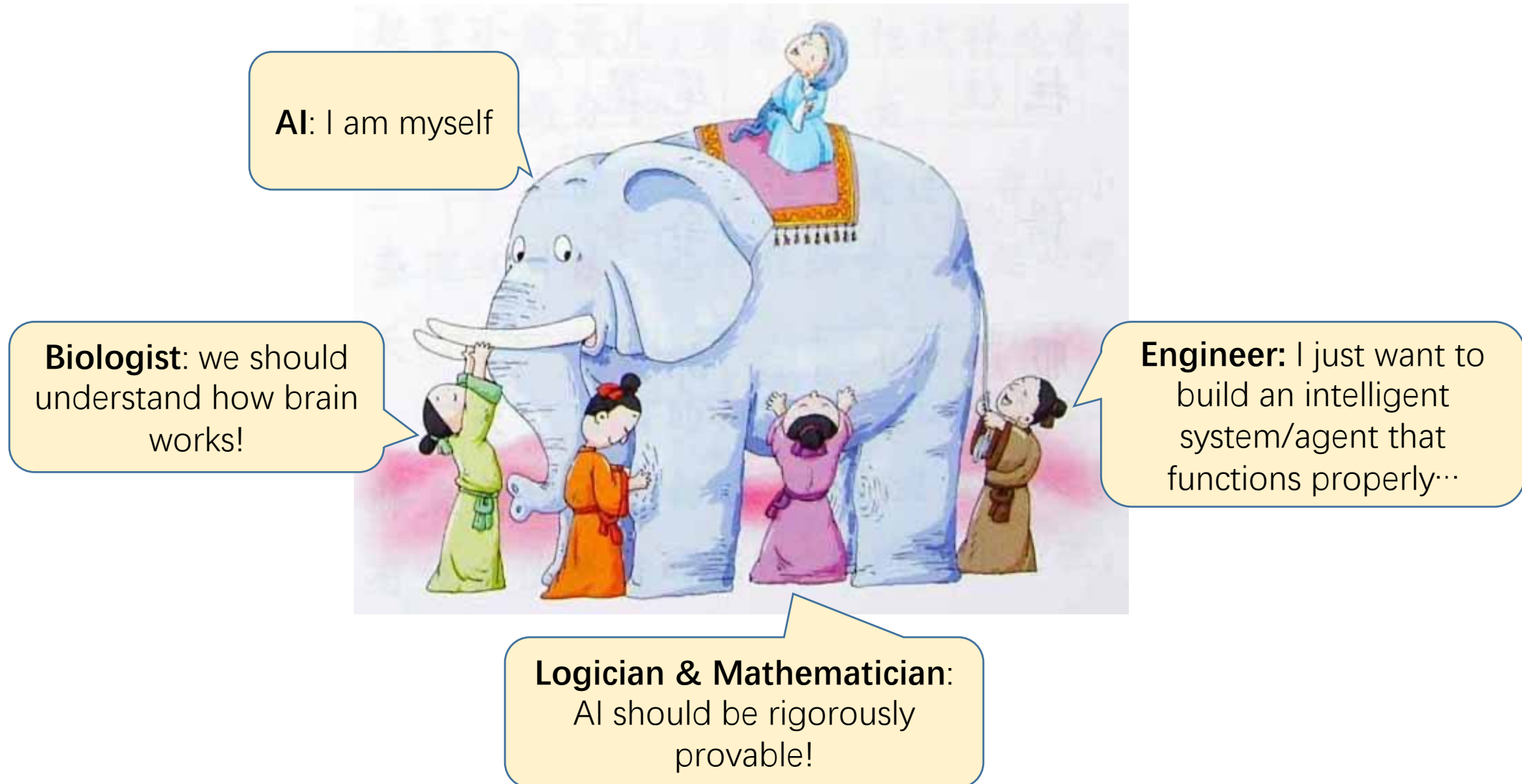
Acting Rationally

“Computational Intelligence: The study of the design of intelligent behavior” (et al., 1998)

“AI ... is concerned with the design of behavior in artifacts.” (Winston, 1992)



Everybody could say something about AI



Everybody could say something about AI



Thinking like a human?

- Intelligent Search
- Machine Learning
- Logical Reasoning

Listening like a human?

- Speech Recognition
- Machine Translation

Seeing like a human?

- Machine Vision
- Autonomous Driving

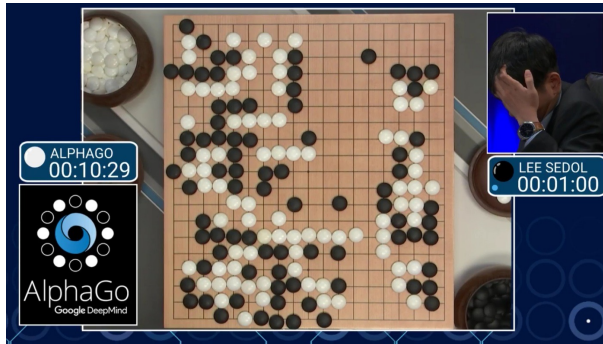
Acting like a human?

- Walking Control

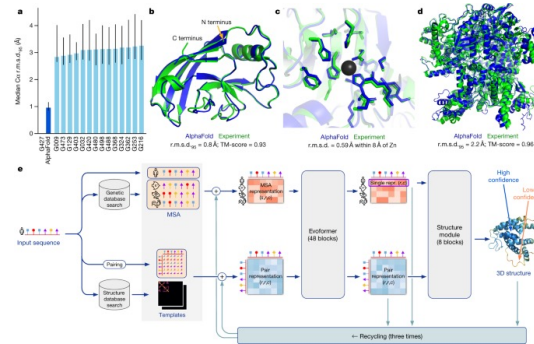
But why AI is important is clear

Powerful (computing) systems have been developed, which have showed comparable or even stronger “intelligence” than human.

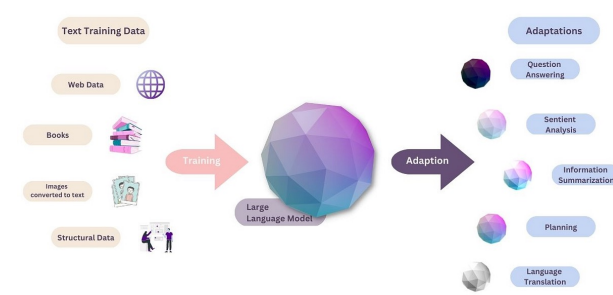
AlphaGo



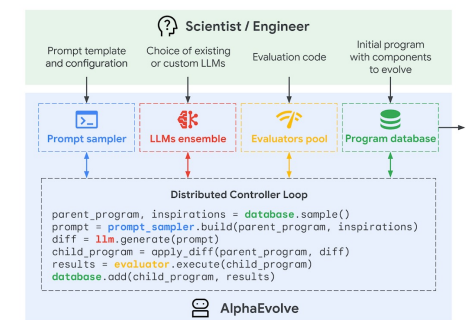
AlphaFold (AI for Science)



Large Language Model



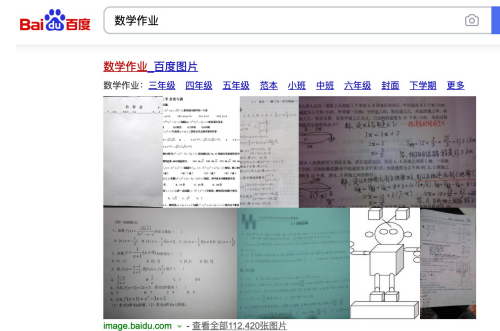
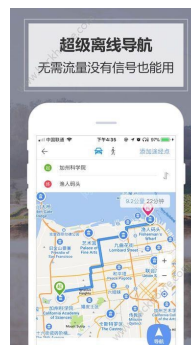
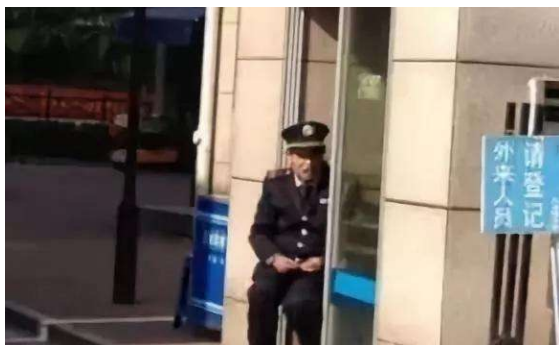
AlphaEvolve



These systems function *like* the brains of human.

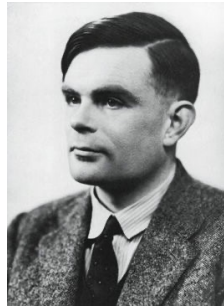
Why is AI **hot** again?

- Making our life easier (more convenient)

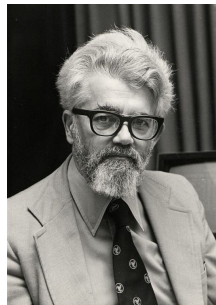


What is AI? – A Historical Perspective

- The rough idea of AI can be dated back to 1950, by **Alan Turing** in his paper “**Computing machinery and intelligence**. Mind, 49:433-460, 1950.”

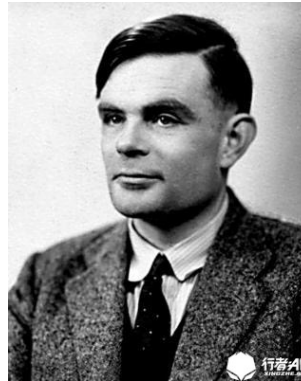
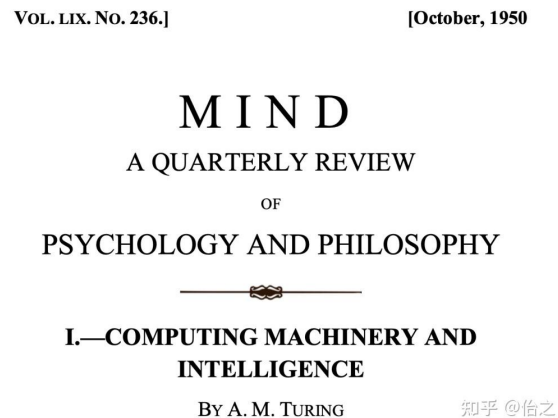


- The **Dartmouth Artificial Intelligence (AI) Conference in 1956**, initiated by **John McCarthy** gave the birth of the AI area.

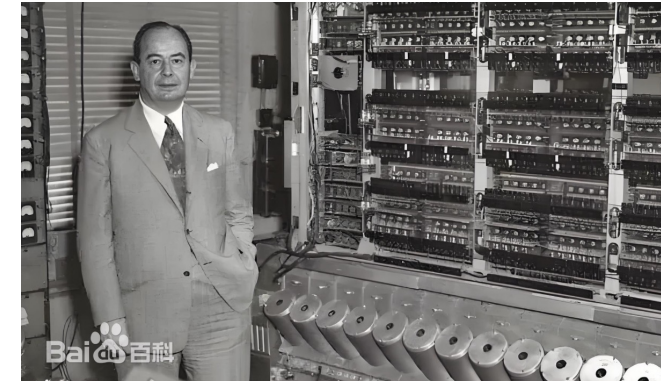


What is AI? – A Historical Perspective

“Computing Machinery and Intelligence”(1950)



Alan Turing (1912-1954)



John von Neumann (1903-1957) & EDVAC in 1950

“Can machines think? ”

- “Consider a particular kind of machine, usually called an “electronic computer” or “digital computer.””
- “The “digital computer” considered is universal.”

What is AI? – A Historical Perspective

Summer Research Project on **Artificial Intelligence** in 1956

IBM701, the first commercial computer in the world (developed in 1952)



[John MacCarthy \(1927-2011\)](#)

Marvin Minsky (1927-2016)

[Claude Shannon \(1916-2001\)](#)

Ray Solomonoff (1926-2009)

Alan Newell (1927-1992)

Herbert Simon (1916-2001)

[Arthur Samuel \(1901-1990\)](#)

Oliver Selfridge (1926-2008)

[Nathaniel Rochester \(1919-2001\)](#)

Trenchard More (1930-2019)



IBM701 Computer

Basically, AI is about how to make machines (e.g., **computing systems**) handle intelligent tasks

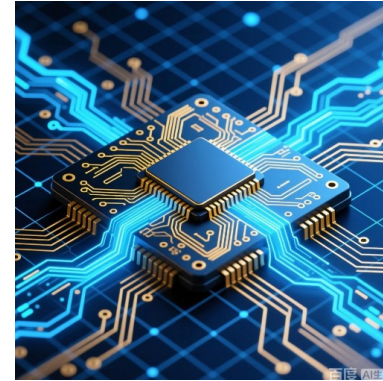
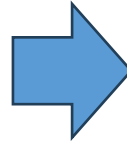
The development of AI is shaped by computers

What is the **key difference** between the intelligence between human and that of the other animals?

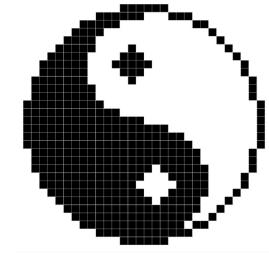
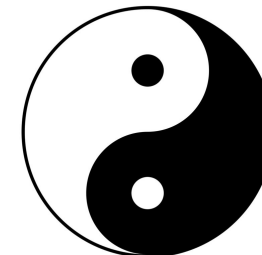
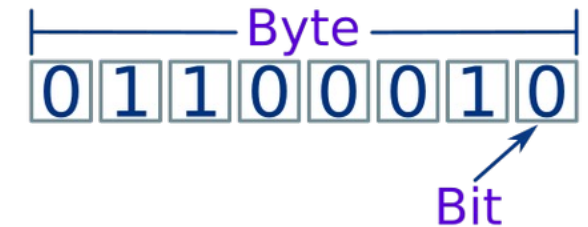


Abstraction → “Language”

The development of AI is shaped by computers



- The way we storage and process information
 - From analog to digital
 - From ambiguous to precise
 - From continuous to discrete

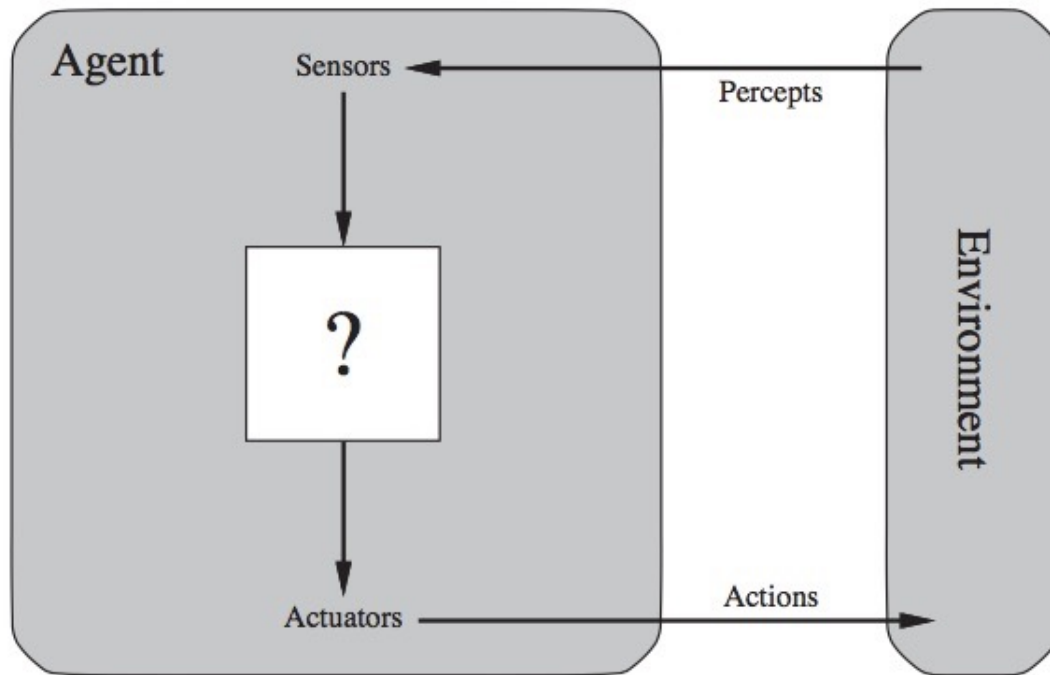


So what's THIS course about?

- We take the engineering perspective.
- We concern building **computing systems** for **applications** that needs **some level of intelligence**.
 - The algorithmic/software level is particularly emphasized

What's THIS course about?

- The term “**Agent**” might be the very first concept for AI.



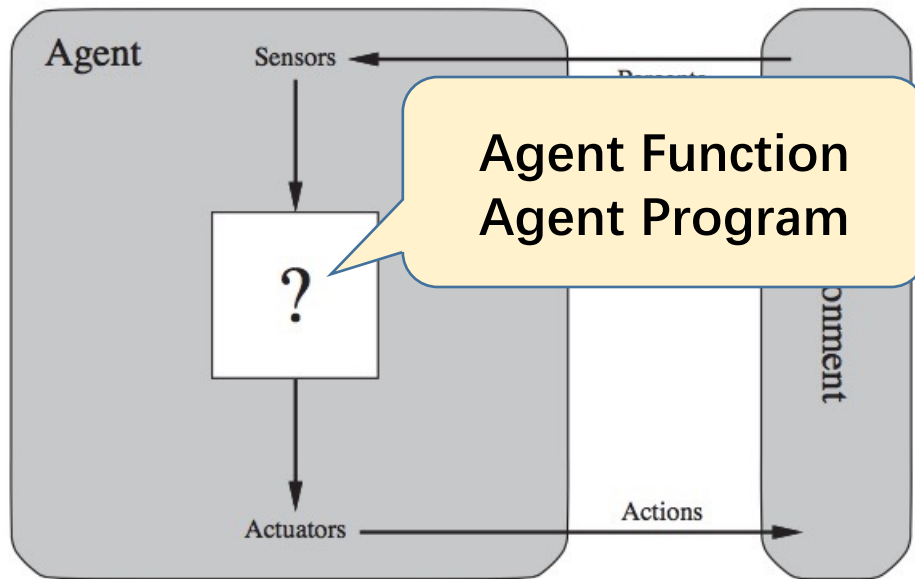
For human

- Sensors: eye, ears
- Actuators: hand/leg...
- ?: brain

What's THIS course about?

- Agent is an abstract concept, it can be everything, similar to a point (object) in a high school physics textbook.
- Agent is the most basic terminology, as well as the entity to investigate, in many classical AI literature.
- Recently, different sub-areas of AI have started using more domain-specific terminology, rather than agent, e.g., “*learner*” in machine learning.

What's THIS course about?

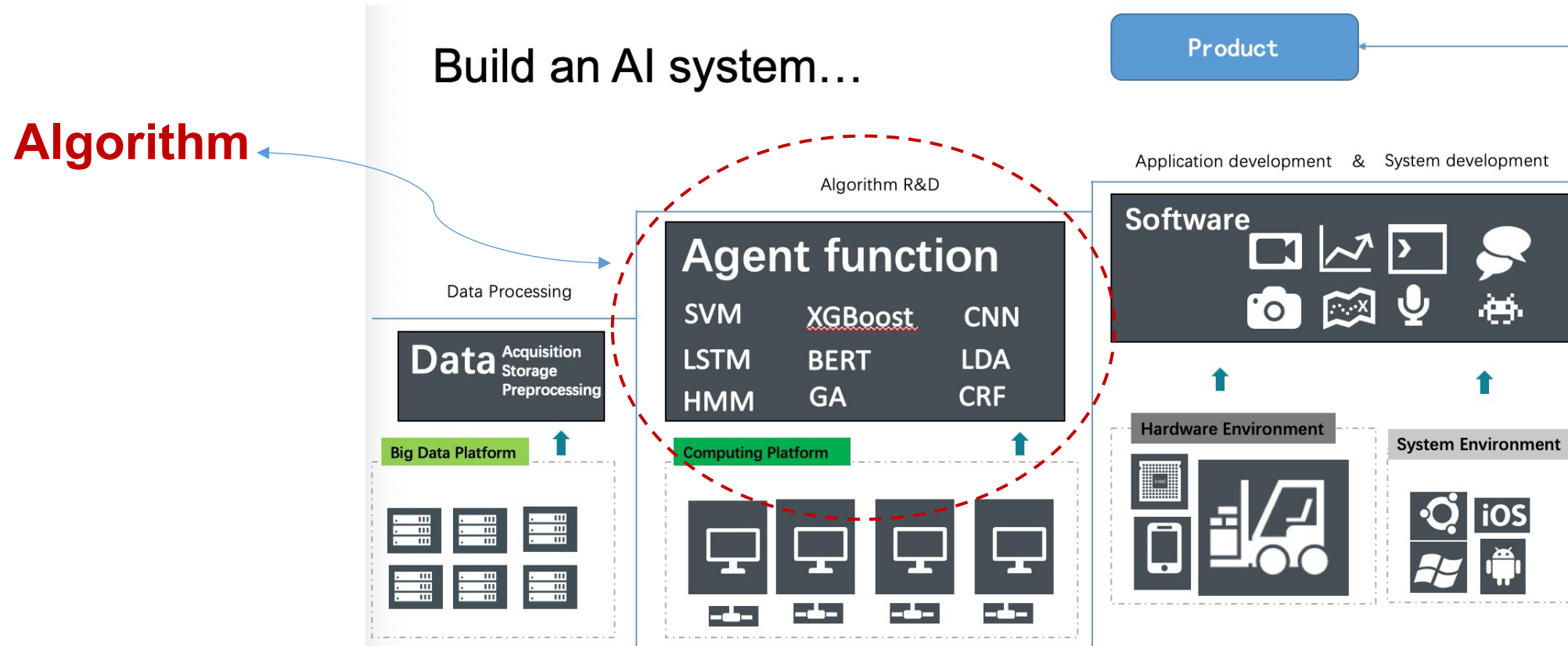


$$F_9(\mathbf{x}) = \sum_{i=1}^D (z_i^2 - 10 \cos(2\pi z_i) + 10) \quad \text{Function}$$

- From the CS viewpoint, an agent's behavior is **described (mathematically)** by the agent function that maps any given percept sequence to an action.

What's THIS course about?

- In a nutshell, this course introduce **how to design various agent functions.**



Course Structure

Lectures: 3 sections

- Problem Solving: AI as *search*
- Learning: *gain experience/knowledge* from data
- Knowledge and Reasoning: represent human knowledge *logically*.

Projects: 3 near-industry/academia-level projects

To be continued